MemeScreen

App schematic. TradingView for memecoins, paper trading only.

Project	MemeScreen, team class project
Document date	October 15, 2026
Build state	Web app and phone version running. Backend is Back4App. iPhone build running through Expo Go over the hotspot. Screener, shared competitions and the legal pages added in this build. Desktop wrapper in the repo.
Rule	Paper trading only. No real funds. No wallet connection.

1. What the app is

MemeScreen charts, grades and paper trades Solana memecoins with live market data. TradingView has the charts but cannot paper trade a memecoin. MemeScreen does both, with a rug-risk grade on every token.

Every new account starts with $10,000 of simulated money. Orders fill with the same costs a real swap would have: a platform fee, a liquidity pool fee, a network fee, and price impact based on the size of the pool. Nothing in the app touches a real wallet.

2. Who it is for

- People who already trade memecoins and want to test an idea without paying for the lesson.
- People who are curious about memecoins and want to learn how fees, slippage and thin liquidity work before they put money in.
- TradingView users who want the same drawing tools on tokens TradingView does not list.
- Students and classes in fintech or trading courses that need a sandbox with a leaderboard.
- Trading communities that want to run their own weekly competition.

3. Brand

Logo: a deadpan coin. An outlined coin face with square eyes, one raised eyebrow and a smirk, drawn in the brand green on near-black. The wordmark is "MemeScreen" with "Screen" in the accent colour.

	Hex	Name	Used for
	#22E4A0	Brand green	Logo, live status dot, grade A, positive states
	#3B82F6	Accent blue	Primary buttons, active tab, links, selected items
	#0C0E12	Background	App background
	#14161C	Surface	Header, cards, panels
	#1A1D25	Raised	Inputs, hover, selected rows

	Hex	Name	Used for
	#242830	Border	All dividers and outlines
	#E4E6EB	Text primary	Main text
	#8A8F9C	Text secondary	Labels and hints
	#26A69A	Candle up	Up candles, long button, take-profit line
	#EF5350	Candle down	Down candles, short button, stop-loss line
	#2962FF	Drawing blue	Chart drawing tools

	Grade	Meaning
	A	Safe
	B	Low risk
	C	Caution
	D	High risk
	F	Likely rug

Type: the system sans-serif font on each device for text. A monospace font with tabular figures for every price, balance and percentage so numbers line up.

4. Platforms

Platform	How it runs	State
Web (any browser)	Vite app served at localhost:5173 in development, static build for hosting	Running
Windows	Tauri 2 wrapper in src-tauri builds an installable .exe from the same code	Wrapper is in the repo. Not built or tested in this session.
macOS	Same Tauri 2 wrapper builds a .app / .dmg. Has to be built on a Mac.	Not built yet
iOS	Expo shell (a React Native WebView app opened with Expo Go) or Safari, Add to Home Screen. Both load the phone app from the laptop over the iPhone hotspot. An EAS build is the path to the App Store and needs an Apple Developer account.	Opening on the iPhone through Expo Go. Screen lock fix waiting on a retest.
Android	Installable web app from Chrome (manifest and service worker are in place). Same wrapper path for the Play Store.	Phone layout running in the browser and in the phone simulator
Join page	join.html shows the QR codes for phones: one per Expo account, plus a browser code	Running
Phone simulator	phone.html shows a phone with a home screen, an app icon, the app, and phone-style notifications	Running

Joining from phones with the QR code

One script on the laptop starts the website and Expo, then opens a join page (join.html) with the QR codes. Several phones can join at once.

Code	What it opens	Steps on the phone
Expo Go	exp://{laptop address}:{port}. The Expo shell, with phone notifications. One code per Expo account.	Install Expo Go and sign in. On the join page tap your Expo account name, then scan that code. iPhone scans with the Camera, Android scans inside Expo Go. Create a MemeScreen account.
No install	http://{laptop address}:5173/?m=1. The phone app in Safari or Chrome.	Scan and open the link. Optional: Add to Home Screen. Create a MemeScreen account.

Item	How it works
Expo accounts	Expo Go opens the project for a signed-in phone only when the laptop serves it under that same Expo account. The script asks for up to 4 Expo logins (the main one plus 3) and starts one Expo server per account.
Ports	8081 main account, 8082 to 8084 extra accounts. The join page shows one tab per account.
Passwords	Typed into the script, passed to the Expo login command, not saved. Later runs reuse the saved Expo sessions.
Network	Phones must reach the laptop. Same hotspot or Wi-Fi uses Put-MemeScreen-on-iPhone.bat. Networks that block device-to-device traffic use Put-MemeScreen-on-Phones-ANYWHERE.bat, which puts the website behind a Cloudflare tunnel and Expo behind its own tunnel.
Laptop address	Picked in this order: iPhone hotspot (172.20.10.x), Windows hotspot (192.168.137.1), Wi-Fi.
Data	Logins and paper accounts are in Back4App, so each phone has its own MemeScreen account.

One codebase. The phone version is the same HTML and app.js as the website with a phone layout on top.

5. Tech stack

Layer	What is used
Front end	Vanilla JavaScript (ES modules), HTML, CSS. Vite for the dev server and build.
Charts	KLineChart 9.8 for candles, indicators and drawing overlays. Canvas for the dashboard and equity charts.
Backend	Back4App (hosted Parse server) over its REST API. In use.
Backend, second option	Node and Express with PostgreSQL. JWT sessions, bcrypt password hashes (12 rounds).
Desktop	Tauri 2
Phone	Responsive phone layout, web app manifest, service worker for notifications
Storage	Back4App holds every account, paper account, profile, leaderboard row and competition. The browser keeps a short-lived copy so the app opens instantly and keeps working through a dropped connection.

All backend calls go through backend.js, which has a Back4App and an Express implementation with the same functions.

6. API integration

For each API: what data points it gives, what screen they land on, and how it works with the app.

API	Data points	Screen	How it interacts with the app
DexScreener token boosts and token profiles	Token addresses of the most boosted and newest Solana tokens, boost count, listed socials	Trade (token list), Screener, Dashboard, landing page token count	Called once a minute from the browser. Builds the list of up to 60 tokens the app tracks. Held and watched tokens are always kept in the list. Boosts and socials feed the social momentum score.
DexScreener tokens	Price, liquidity, 24h volume, market cap, buys and sells in the last 5 minutes and last hour, 5m and 1h volume, 1h and 24h change, pair age, pair address	Trade (list, token header, stats, order ticket), Screener, Dashboard (featured token, top movers, fresh launches), News heatmap, Portfolio and header (position values)	Polled every 2 seconds. Feeds the rug-risk grade, the screener columns, fills and price impact, position P&L;, limit and stop triggers, take profit and stop loss, liquidations and price alerts.
Solana JSON-RPC, wallet	SOL balance and every token account of a connected Phantom wallet	Portfolio (Phantom wallet card), Broker window	Called when Phantom connects and every 60 seconds after. Read-only: the app never asks the wallet to sign.
Jupiter swap API	Quote (amount out, price impact, route) and a ready-to-sign swap transaction	Trade, live ticket	Live mode only. Quote on request, swap built when the user presses Sign. lite-api host needs no key; a free key in config.js raises the limit.
Expo push API	Delivery of push notifications to the iPhone app	Settings, Notifications; alerts while the app is closed	The phone registers its Expo push token in Back4App (PushDevice). The sendPush and pushToUser cloud functions post to Expo, which delivers through Apple and Google.
Phantom deep links	Wallet address and a signing session, encrypted to the app's one-time key; signed transactions back	iPhone app: Broker window, live ticket	The Expo shell opens phantom.app/ul/v1/connect with its public key and a link back to itself. Phantom asks the user, then opens that link with the address. The shell decrypts it and hands it to the web app, which returns to the Portfolio page.
RugCheck token report	Holder count, top-10 holder share, insider share, bundled supply, mint and freeze authority, creator address and the creator's earlier tokens, risk list	Screener (Holders, Top-10, Insider, Bundled, Mint, Dev history columns)	One request per token, queued a few at a time, cached 10 minutes. Screens that need a holder figure treat a token as not matching until its report has arrived.

API	Data points	Screen	How it interacts with the app
DexScreener SOL pair	SOL price in USD	Trade (order ticket), Dashboard (Solana tile)	Once a minute. Turns the SOL size buttons into dollars and converts the trade stream into USD prices.
Solana JSON-RPC, pool watch	Reserves of the selected token's pool: pump.fun bonding-curve virtual reserves, PumpSwap pool token accounts, or Raydium AMM v4 vaults	Trade (price, current candle, header), chart source line	The price the chart trades on. Push updates through accountSubscribe on the RPC WebSocket, polled every 1.5 seconds as the fallback (1 second with a Helius key). Every swap changes the reserves, so the candle moves with the chain instead of waiting for DexScreener. Works keyless on the public RPC; a free Helius key in config.js gives a reliable WebSocket and higher limits.
PumpPortal WebSocket	Each on-chain trade: token, buy or sell, SOL amount, token amount	Screener (Whales column), header status line	Trade subscriptions became metered (paid per event) in 2026, so the stream usually delivers nothing without a funded key. Kept for whale prints when it does. The pool watch above replaced it as the price source.
GeckoTerminal OHLCV	300 candles per timeframe: open, high, low, close, volume	Trade (chart), Dashboard (featured chart)	Fetched when a token or timeframe is opened and re-fetched every 30 seconds to realign closed candles. The in-progress candle is built in the app from the pool watch. Indicators and drawing tools sit on top of this data.
CoinGecko simple price and global	BTC, ETH and SOL price with 24h change, total crypto market cap with 24h change	Dashboard (market strip)	Called by the backend, not the browser, and cached 60 seconds. Express route /api/market or Back4App cloud function market.
Alternative.me	Fear and Greed value and label	Dashboard (market strip)	Browser call when the dashboard loads, refreshed every 5 minutes.
cryptocurrency.cv news	Headline, link, source, publish time, 50 articles	Header ticker on every page, Dashboard (latest news), News page	Every 2 minutes. A headline that mentions a watched or held token fires a news alert.
rss2json with publisher RSS feeds	Same fields from Cointelegraph, CoinDesk, Decrypt, CryptoSlate, Bitcoin Magazine, CryptoPotato, BeInCrypto	Same as above	Fallback only, used when the main news feed returns nothing.

API	Data points	Screen	How it interacts with the app
Back4App Parse REST	User, session token, paper accounts, profile (watchlist, alerts, saved screens), leaderboard rows, competitions	Log in and sign up, Settings, header account switcher, Screener, Competitions, Legal (delete account)	Reads everything on login. Writes about 1.2 seconds after any change. Checks for changes from other devices every 30 seconds. Competitions are created, joined and left with direct calls.
MemeScreen Express API	Same records from PostgreSQL	Same as Back4App	Same behaviour. The app uses one or the other depending on the server setting.

None of the market, holder or news APIs need a key. Back4App needs the Application ID and the JavaScript key. The Express API uses the JWT it hands out at login.

Browser and device APIs

API	Used for
Notifications API and Service Worker	System notifications for alerts on desktop and phone
Web App Manifest	Install to home screen on iOS and Android
BroadcastChannel	Keeps two open copies of the app (website and phone simulator) in step
postMessage	The app reports alerts and sign-in to the phone simulator and to the Expo shell
expo-notifications and react-native-webview	Expo shell on the iPhone: shows the app full screen and turns its alerts into iOS notifications
WebSocket	PumpPortal trade stream
Web Storage	Short-lived copy of the signed-in state so the app opens instantly

Outbound links only, no data pulled

DexScreener token page, X search for the ticker, Google News search, Pump.fun, Jupiter, Birdeye.

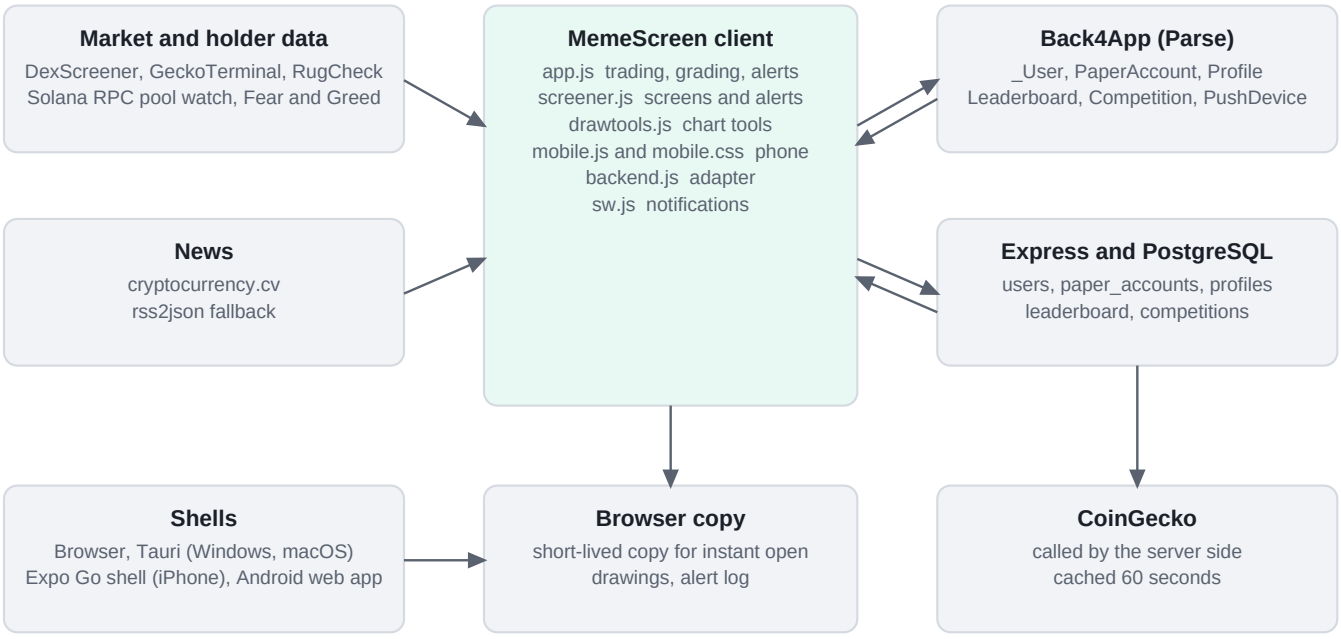
7. Storage: Back4App

User data is stored in Back4App. Every device the user signs in on reads the same records. The browser keeps only a short-lived copy so the app opens instantly and keeps working if the connection drops for a moment.

Data	Back4App class	Notes
Account (email, nickname, password hash)	_User	The password is stored only as a hash
Session	Session token issued at login	Kept while "keep me signed in" is ticked, otherwise ends when the browser closes
Paper accounts list	PaperAccount rows (name, size)	Up to 25 per user

Data	Back4App class	Notes
Paper account state: cash, positions, trades, orders, equity log	PaperAccount.state	Written about 1.2 seconds after any change. Back4App wins on login.
Watchlist, alert rules, indicator settings, active account, saved screens	Profile.data	Same write rule. The wallet is never stored: a Phantom connection lives in memory for one session.
Leaderboard	Leaderboard rows	Main account only
User-created competitions	Competition rows (name, code, owner, members)	Anyone can read. Join and leave add or remove the user from the member list.
Alert log	Not stored	Kept in the browser for the session
Chart drawings	Not stored yet	Kept in the browser. Planned to move to Back4App.
RugCheck reports	Not stored	Held in the page for 10 minutes
Push devices	PushDevice rows (owner, Expo push token, platform)	One row per phone. Owner-only ACL. Written when the user says yes to notifications.
Account deletion	Deletes the _User row and every PaperAccount, Profile, Leaderboard and PushDevice row the user owns	Settings, Legal, Delete my account and data. The browser copy is cleared too.
Guest mode	Nothing	Nothing is sent to the server

8. System schematic



Prices, holder data and news go from the public APIs straight to the browser. Back4App stores users, paper accounts, profiles, leaderboard rows and competitions. Trade math and screening run in the client. The Expo shell on the iPhone shows the same client full screen and turns its alerts into iOS notifications.

Data model

Record	Fields	Who can read and write
User	email, nickname, password hash	The user
PaperAccount	owner, name, starting size, state (cash, positions, trades, open orders, equity log, week start equity, reset time)	Owner only
Profile	owner, watchlist, alert rules, indicator settings, last active account, saved screens	Owner only
Leaderboard	owner, period (week or all), week id, nickname, return, average grade, trades, score	Everyone reads, owner writes
Competition	name, 6-character join code, owner, owner nickname, member nicknames, member ids	Everyone reads. Members add and remove themselves.
PushDevice	owner, Expo push token, platform	Owner only

A user can have up to 25 paper accounts, each with its own starting balance. The account called main is the one that competes on the leaderboard. The others are private sandboxes.

Sign-in and sessions

- Email and password. Passwords are stored only as a bcrypt hash on both backends.
- "Keep me signed in" ticked: the session survives closing the browser.
- Unticked: the session ends when the browser closes.
- Guest mode: everything works and saves to the browser only. No leaderboard.
- The Express server limits sign-in attempts to 20 per 10 minutes per IP address.

Sync

- Every change saves to the device at once, then to the backend about 1.2 seconds later. Failed saves retry every 10 seconds.
- On login the backend copy wins over the device cache.
- While idle the app checks the backend every 30 seconds for changes made on another device.
- Two copies open in the same browser update each other instantly.
- If two devices edit the same account at the same moment, the last save wins.

9. Trading engine

Item	Rule
Starting balance	$10,000 on main. Any amount from $1 on extra accounts.
Order types	Market, limit, stop. Long and short.
Exits	Take profit and stop loss, typed as dollars of profit or risk and converted to a trigger price for the side being traded
Order sizing	SOL presets (0.5, 1, 2, 5, 10), percent of cash (10, 25, 50, 100), or a custom dollar amount
Platform fee	1.0%
Liquidity pool fee	0.3%
Network fee	$0.10 per order
Price impact	order size divided by (half the pool liquidity plus order size)
Warnings	Confirmation needed when impact is over 25% or the token is graded D or F
Shorts	Liquidated when the position value reaches zero
Chart orders	Right-click the chart (long-press on a phone) to place a limit, stop, take profit or stop loss at that price
Leaderboard score	return x (0.5 + average entry grade / 4 x 0.7). Trading safer tokens scores higher.

Rug-risk grade, 0 to 100

Input	Points
Liquidity, full marks at $50,000	35
Buy and sell balance over the last hour	15
24h volume compared to liquidity (over 20x looks like wash trading)	15
Pair age, full marks at 24 hours	20
Market cap compared to liquidity	15
Cap	Any token with under $5,000 liquidity cannot score above 30
Letter	A 85 and up, B 70, C 55, D 40, F below 40

10. Pages, web application

Page	What is on it
Landing	Headline, product preview, six feature cards, charting section, how it works, sign-up buttons, guest entry
Log in / Create account	Email, password with show and hide, nickname on sign-up, keep me signed in, forgot password, browse as guest, server status with a link to change the backend
Forgot password	Email only. Back4App sends the reset email with its own link and page. On the Express server the link is printed in the server console and opens the app on a Choose a new password screen; the link lasts one hour and works once.
Dashboard	Featured token with chart and stats, top movers, crypto market cap, Fear and Greed, Bitcoin, Solana, account equity, fresh B-or-better launches, latest headlines. The "Market summary" heading is a button that opens the featured token in Trade.
Screener	Bloomberg-style table of every tracked token: ticker, grade, age, market cap, liquidity, 5m volume, 1h volume, buy/sell ratio, holders, top-10 share, insider share, bundled supply, mint authority, dev history, whale trades, social momentum. Sortable columns. Preset screens, user-defined screens with min and max on every column, saved to the account, optional alert when a token starts matching. Tap a row to chart it.
Trade	Token list with search and filters (grade, liquidity, age, sort, watchlist). Token header with grade and red flags. Chart with timeframes, indicators, drawing toolbar, full screen. Order ticket. Bottom panel with positions, open orders and trade history. Size presets switch to $5 to $100 when the account is under $2,500.
Trade, live mode	When the account switcher is on the Phantom wallet the ticket turns green: Buy (■0.01 to ■0.5 or custom) or Sell (25, 50, 100% of the holding), Get Jupiter quote, then Sign with Phantom. Test mode is the default: Phantom signs the real transaction and nothing is sent. Ticking Send for real broadcasts it. The positions tab shows the wallet's real holdings with a Sell button each.
News	Market heatmap of tracked tokens, headline feed, filter by token or keyword, quick links to chart, X and web search for a token
Portfolio	Equity, total P&L;, win rate, fees paid, open positions, this week. Equity curve. P&L; by grade at entry. Full trade history.
Competitions	Your weekly and all-time standing, official boards, user-created competitions: create one and get a code, type a code to join, a board of just that competition's members for the week, leave. Outside competitions as links.
Settings	Account, paper trading accounts (create, switch, delete, reset), connect a broker, notifications and alert rules, recent alerts, fee model, server, legal
Legal	Privacy policy, terms of use, refunds (no purchases exist), cookies and site data, accessibility statement, and a Delete my account and data button. Linked from Settings, the sign-up form and the landing page footer.
Broker window	Paper Trading shows connected. Phantom connects for real; once connected the card offers Switch to live wallet, Back to paper and Disconnect: on a laptop through the Phantom extension, on a phone by opening the site inside the Phantom app. Coinbase, Kraken, OKX, Jupiter, Axiom, Photon, GMGN, BullX, Birdeye and Crypto.com stay greyed out and fail with a message.

Page	What is on it
Phantom wallet card (Portfolio)	Real SOL balance, every token the wallet holds with amount, DexScreener price, value, liquidity and the rug-risk grade, wallet total, Chart button per token, Refresh and Disconnect. Read-only: orders in the app stay paper.
Header, every page	Logo, navigation, account switcher (paper accounts and the Phantom wallet), + Deposit, broker button, equity, P&L;, this week, user name, log out, live data status, sync status, scrolling news ticker
Deposit	Paper accounts: any amount of paper money in or out of the active account, so small sizes can be practised. Wallet: shows the Phantom address and a QR code to send SOL to.
Notifications ask	Once per login a card asks whether to turn notifications on. Yes asks the browser, or on the iPhone app the phone, and registers the push token.

11. Pages, phone application

Same pages and the same features as the website. The differences are in layout.

Screen	What is on it
Home screen (simulator only)	Clock widget, MemeScreen icon with an unread badge, placeholder app icons, dock. Tap the icon to launch.
Sign in	The first screen the app opens on. Log in, create account, keep me signed in, browse as guest.
Tab bar	Home, Screen, Trade, News, Portfolio, Compete, Settings, fixed at the bottom
Home	Dashboard in one column
Screen	The screener table scrolls sideways with the ticker column pinned. The screen editor opens as a bottom sheet.
Trade	Three segments: Markets, Chart and trade, Positions. Picking a token jumps to its chart. A floating Trade button scrolls to the order ticket.
Chart	The chart owns every gesture, as in the TradingView app: one finger pans in both directions, two fingers zoom, the page does not scroll underneath. Drawing toolbar runs across the top. Tool lists open as a bottom sheet. Tap to place points. Long-press for the order menu. Full-screen mode.
Positions	Each position is a card with P&L;, quantity, entry, current price, value, TP and SL, and 25%, 50% and Close buttons
News, Portfolio, Compete	Single-column versions of the web pages. News links open in Safari, not inside the app.
Settings	Same as web. The user name and log out button live here because the header has no room.
Phantom on the iPhone app	Broker window, Phantom, Connect. The Expo shell opens Phantom with a deep link; after approval Phantom returns to the app and the Portfolio page shows the wallet. Read-only.

Screen	What is on it
iPhone through Expo Go	The Expo shell shows the phone app full screen with the status bar and home indicator respected. It loads the app from the laptop over the hotspot (or a tunnel), only MemeScreen pages can load inside it, and every outside link goes to Safari. Alerts arrive as iOS notifications and tapping one opens that token's chart. Sideways drag, bounce, pinch zoom and swipe-back are off.
Joining	Phone hotspot on, laptop on it, run Put-MemeScreen-on-iPhone.bat, scan the code on the join page with the Camera. One Expo Go code per Expo account, plus a browser code for phones without Expo Go.
Notifications	A banner drops from the top. In the simulator, alerts arrive on the home screen while the app is in the background, badge the icon, and collect in a notification centre opened from the status bar. Tapping one opens that token's chart.

12. Chart tools

Group	Tools
Cursors	Cross, dot, arrow, eraser
Trend lines	Trend line, ray, info line, extended line, arrow, horizontal line, horizontal ray, vertical line, parallel channel, price channel
Gann, Fib and pitchfork	Fib retracement, trend-based Fib extension, pitchfork, Schiff pitchfork, Gann box
Shapes	Rectangle, circle, triangle, parallelogram
Annotations	Text, callout, price note, arrow marker up, arrow marker down, flag
Patterns	XABCD, head and shoulders, Elliott impulse 12345, Elliott correction ABC, triangle pattern
Prediction and measurement	Long position, short position (with reward to risk), price range, date range, date and price range
Controls	Magnet, lock all, hide all, delete selected, clear all
Indicators	MA 20, MA 50, EMA 20, Bollinger Bands, RSI 14, volume
Timeframes	1m, 5m, 15m, 1h

The live candle moves with the chain: the app watches the selected token's pool reserves on Solana (push, with 1.5 second polling as the fallback) and updates the candle on every swap. Closed candles are realigned to GeckoTerminal every 30 seconds without resetting the view. The source line under the chart shows which pool type is being watched and how old the last update is. Drawings save per token in the browser and come back after a reload or a timeframe change. The chart axis is market cap. Entry, take-profit, stop-loss and resting order lines are labelled with price and dollar amount, and pin to the edge of the chart when the level is out of view.

13. Alerts

Alert	Default
Price move in 1 minute	5%, on
Price move in 5 minutes	15%, on
Safety grade downgrade	On
Short position liquidated	On
New pair launch at a minimum grade	B or better, on
Limit or stop order filled	Always
Take profit or stop loss hit	Always
News headline mentions a watched or held token	Always
A token starts matching a saved screen	Off until "Alert me" is ticked on that screen

Alerts watch starred tokens, open positions and saved screens. The same alert does not repeat for 5 minutes. Each one shows as a toast, an on-screen card, an entry in the alert log, and a system notification if the user has allowed them. Screen alerts wait 90 seconds after the app opens so the first load does not fire a burst.

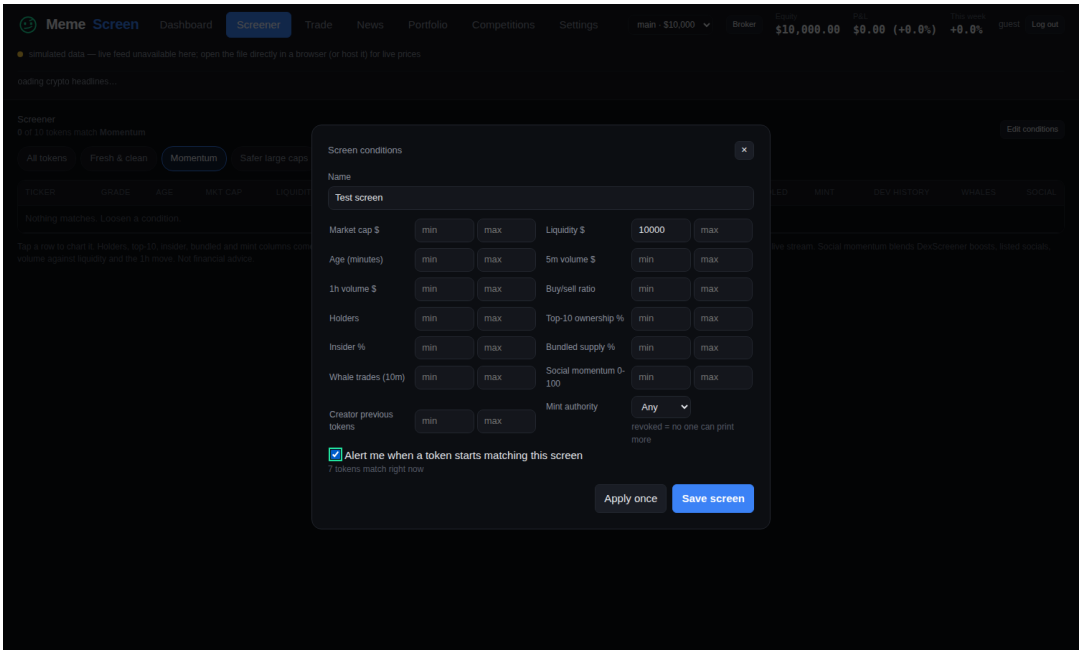
14. Screener

The screener is a table of every tracked token with the numbers a memecoin trader checks before entering, and screens that cut the table down to the tokens that fit a rule.

Column	Source	What it means
Age	DexScreener	Time since the pair was created
Market cap, Liquidity	DexScreener	Fully diluted value and pool liquidity in dollars
5m vol, 1h vol	DexScreener	Dollar volume in the last 5 minutes and last hour
Buy/sell	DexScreener	Buys divided by sells in the last 5 minutes
Holders	RugCheck	Number of wallets holding the token
Top-10	RugCheck	Share of supply held by the ten largest wallets
Insider	RugCheck	Share of supply held by wallets RugCheck flags as insiders
Bundled	RugCheck	Share of supply bought in the same block as launch. Falls back to the insider share when RugCheck reports no bundle.
Mint	RugCheck	Revoked (nobody can print more) or live
Dev history	RugCheck	How many tokens the creator wallet launched before this one
Whales	PumpPortal stream	Trades of $1,000 or more in the last 10 minutes
Social	Worked out in the app	0 to 100. Blends DexScreener boosts, listed socials, volume against liquidity and the 1h move

Preset	Conditions
Fresh & clean	Market cap $250K to $5M, liquidity over $100K, under 24 hours old, 1h volume over $500K, top-10 under 20%, bundled under 10%, mint revoked, creator has launched nothing before
Momentum	1h volume over $250K, 5m volume over $25K, buy/sell over 1.2, at least one whale trade
Safer large caps	Market cap over $1M, liquidity over $250K, top-10 under 30%, mint revoked

A user-defined screen has a name, a min and max on every column, a mint authority choice and an alert switch. Saving it adds a chip next to the presets and stores it on the account, so it appears on every device the user signs in on. A token with an unknown value never passes a numeric condition, so a screen that filters on holders does not show a token whose RugCheck report has not arrived.



The screen editor on the web. On the phone the same editor opens as a bottom sheet.

15. Legal and compliance

Item	State	Where
Privacy policy	Yes	Settings, Legal. What is stored (email, nickname, password hash, paper accounts, profile), where (Back4App, US), what never leaves the device, third-party APIs called from the browser
Terms of use	Yes	Paper trading only, no financial advice, no real funds, account rules
Refund policy	Yes	Nothing is sold, so there is nothing to refund. Stated so the page is complete.
Cookies and site data	Yes	No cookies. States what the browser keeps for the session and that user data lives in Back4App.
Consent on sign-up	Yes	The create-account form needs the "I agree to the terms and privacy policy" box ticked
Data deletion request	Yes	Delete my account and data button. Asks for confirmation, then removes the user and every row they own from the backend and every MemeScreen key from the device.
Accessibility: alt text	Yes	Icons carry aria-labels, decorative images are hidden from screen readers
Accessibility: contrast	Yes	Body text on the background is 15:1. Secondary text is 5.6:1. Both above the 4.5:1 minimum.
Accessibility: keyboard	Yes	Every table row, chip, tab and button is reachable with Tab and activated with Enter or Space. The screen editor opens on the name field and closes with Escape.
Apple guidelines on the phone	Yes	All tap targets at least 44 points. Safe-area insets. Content stays inside the screen with no sideways scroll. Reduced motion respected. Outside links leave the app.
Age	Stated	Terms ask for users 18 and over. Not verified.

16. QA: feature list against what is in the app

Feature	State	What is there now
Accounts: sign up, log in, log out	Yes	Email and password, bcrypt hashes, each user isolated
Keep me signed in	Yes	Web and phone
Forgot password	Yes, needs a live check	Back4App sends the email. Tested against the Express server end to end; the Back4App email itself is unconfirmed.
Guest mode	Yes	Nothing is sent to the server
Multiple paper trading accounts per user	Yes	Create, switch, delete, reset, any starting balance, synced to the backend
Back4App backend	Yes	Connected with the team's Application ID and JavaScript key. Accounts, paper accounts, profile and leaderboard are stored in Back4App.
Express and PostgreSQL backend	Yes	Kept as a second option. Not in use now.
Token list filters	Yes	Search, grade, liquidity, age, sort, watchlist
Bloomberg-style screener	Yes	16 columns, sortable, three presets, user-defined screens with min and max on every column
Saved screens and screen alerts	Yes	Saved to the account. Alert fires when a token starts matching.
Holder, insider, bundle and creator data	Yes, needs a live check	From RugCheck. Tested with simulated tokens only, so the live columns are unconfirmed.
Rug-risk grade A to F with red flags	Yes	
Live prices	Yes, needs a live check	Selected token from its pool on Solana (push and 1.5 second polling); the rest from DexScreener every 2 seconds. Tested with a simulated pool only.
Candlestick chart, timeframes, indicators	Yes	1m 5m 15m 1h. MA, EMA, Bollinger, RSI, volume.
TradingView-style drawing tools	Yes	39 tools in 7 groups, saved per token
Market, limit and stop orders, long and short	Yes	
Take profit and stop loss	Yes	Set in dollars, shown as labelled lines on the chart
Real fees, slippage and price impact	Yes	
Positions, open orders, trade history	Yes	
Portfolio journal	Yes	Equity curve, win rate, fees, P&L; by grade
Weekly and all-time leaderboard	Yes	Main account only
User-created competitions	Yes	Stored in Back4App. Create, join by code, per-competition board, leave.
News feed, ticker, per-token news, heatmap	Yes	Links open in a new tab, or in Safari from the iPhone app

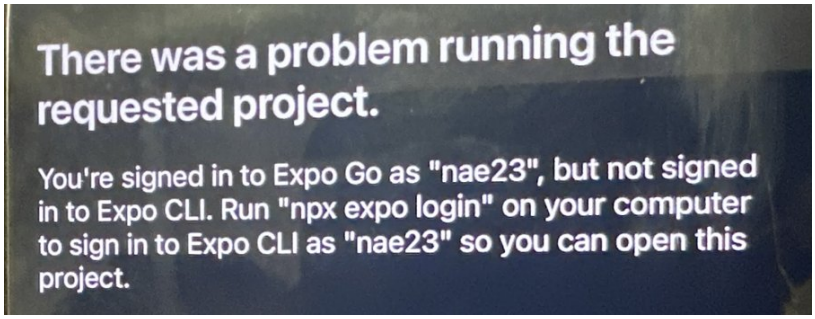
Feature	State	What is there now
Privacy policy, terms, refunds, cookies, accessibility statement	Yes	Legal page
Consent on sign-up	Yes	Required checkbox
Delete account and data	Yes	Removes the user and their rows from the backend and clears the device
Price, grade, launch, order and news alerts	Yes	While the app is open
Push notifications with the app closed	Yes, needs a live check	iPhone app registers its Expo push token; sendPush and pushToUser cloud functions deliver through Expo. Needs the cloud code deployed and a test from a real phone.
Phantom wallet	Yes, needs a live check	Connect, real holdings priced and graded. Session only: closing the app, reloading, or 10 minutes in the background disconnects it. Tested with a simulated wallet and RPC only.
Switch between paper and live wallet	Yes	Account switcher in the header, Broker window buttons
Live orders through Jupiter and Phantom	Yes, needs a live check	Quote, build, sign. Test mode signs without sending; Send for real broadcasts. Tested with a simulated Jupiter and Phantom.
Deposit and withdraw paper money	Yes	Any amount; size presets adapt to small accounts
Notifications prompt at login	Yes	Once per login, Settings can change it
Other broker connections	Placeholder	Listed greyed out. A connection attempt runs and fails with a message.
Real wallet trading	Yes, behind Send for real	Off by default; test mode signs without broadcasting.
Token sentiment panel (bull or bear)	No	Planned. The red-flag list on each token is the closest thing today.
Custom indicator and strategy builder with terminal	No	Planned
AI strategy testing	No	Planned
Phone layout with every web feature	Yes	
Phone simulator with home screen and notifications	Yes	
Runs on an iPhone	Partial	Through Expo Go or Safari Add to Home Screen, served from the laptop. No App Store build.
Several phones at once	Yes	Join page with one Expo Go code per Expo account (up to 4) and a browser code for everyone else.
Runs on Android	Partial	Same routes. Not tested on a device.

Feature	State	What is there now
Windows and macOS installers	Partial	Tauri wrapper is in the repo. Not built.

17. QA: what is broken

Open

1. App moves around inside Expo on the iPhone	Fix written, needs retest on the phone
Description	In Expo Go the app is not locked to the screen. It can be dragged sideways, bounces past its edges and can be pinch-zoomed, so it does not feel like a native app.
Expected behaviour	The app fills the screen and stays put. Only the content scrolls, up and down. No sideways movement, no bounce, no zoom. The page is now fixed in place on phones, with one vertical scroll area inside it, zoom is turned off, and the Expo WebView has bounce, pinch and swipe-back disabled.
Image proof	No capture yet

2. Expo Go refuses to open the project	Fix written, needs retest
Description	A phone signed in to Expo Go gets an account error when the laptop is not serving the project under that same Expo account. A phone with no Expo account fails when the laptop is signed in.
Expected behaviour	Every phone that scans its code opens MemeScreen. The script now runs one Expo server per Expo account and the join page gives each account its own code.
Image proof	

3. Accounts from the old server do not exist in Back4App	Open
Description	The app now uses Back4App. Logins created on the Express server are in a different database, so they are rejected.
Expected behaviour	Each user creates a new account once. After that the same login works on the website, the phone simulator and the iPhone.
Image proof	No capture yet

4. Alerts stop when the app is closed	**Half fixed**
Description	Alerts are worked out inside the app from live prices. With the app or tab closed nothing is watching. The delivery side now exists: push tokens are stored and the pushToUser cloud function sends. Nothing calls it yet.
Expected behaviour	A scheduled cloud job evaluates watched tokens and saved screens on the server and calls pushToUser, so a phone gets the alert with the app closed.
Image proof	No capture yet

5. No system notifications on a phone over plain http	**Open**
Description	Browsers only allow system notifications on localhost or https. A phone that opens the laptop address over Wi-Fi gets the in-app banner only. The Expo shell avoids this by using iOS notifications directly.
Expected behaviour	Hosted on https, the installed web app shows system notifications on the phone.
Image proof	No capture yet

6. Drawings stay on one device	**Open**
Description	Chart drawings are kept in the browser, so they do not follow the login to another device.
Expected behaviour	Drawings are saved to Back4App and appear wherever the user signs in.
Image proof	No capture yet

7. Second chart point ignored on a fast click	**Open**
Description	The chart library reads two clicks less than half a second apart as a double-click, so the second point of a drawing is dropped.
Expected behaviour	Every click places a point no matter how fast.
Image proof	No capture yet

8. Live data not checked in the latest build	**Open**
Description	The build machine could not reach DexScreener, PumpPortal, GeckoTerminal, RugCheck or the news feeds, so the latest changes were tested on simulated prices.
Expected behaviour	Status line shows "live" or "streaming", prices move, the chart loads real candles, the screener holder columns fill in. To be confirmed on the team's machine.
Image proof	No capture yet

9. RugCheck may refuse browser requests	**Open**
Description	The holder columns call RugCheck straight from the browser. If RugCheck blocks cross-site requests or rate-limits the page, those columns stay empty and the screens that use them match nothing.
Expected behaviour	Holders, top-10, insider, bundled, mint and dev history fill in for every token within a minute or two. If they do not, the calls move to a Back4App cloud function.
Image proof	No capture yet

10. Seven tabs on the phone	**Open**
Description	Apple recommends up to five tabs in a tab bar. The phone has seven: Home, Screen, Trade, News, Portfolio, Compete, Settings.
Expected behaviour	Five tabs, with Portfolio, Compete and Settings under a More tab.
Image proof	No capture yet

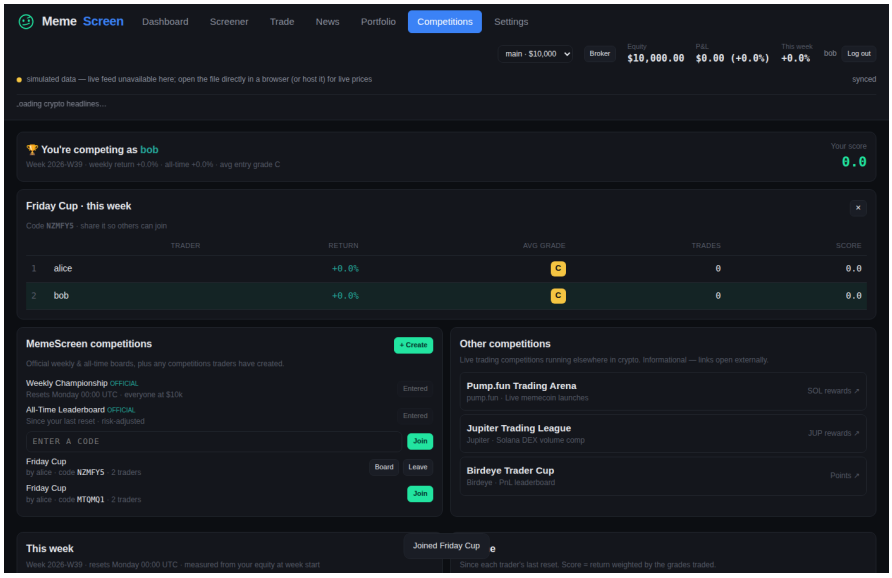
Fixed in this build

11. App jumped to Phantom on every login and the wallet could not be removed	**Fixed**
Description	With a wallet address remembered on the account, the phone app tried a silent reconnect at login, found no wallet in the page and fell into the open-Phantom branch. Disconnect also came back after the next profile sync, and the notifications card never got a chance to show.
Expected behaviour	Login never opens Phantom by itself; only tapping Connect does. The wallet is no longer remembered at all: it is connected for one session and dropped on close, reload, or after 10 minutes in the background. Disconnect Phantom is visible in the Broker window, in Settings and on the wallet card, and sticks for the session even if an old profile copy arrives. The notifications card shows at login as intended.
Image proof	No capture yet

12. Chart and candles lagged behind the market	**Fixed, needs a live check**
Description	The live candle only moved when DexScreener's price field changed, which is often 10 to 60 seconds behind, and PumpPortal's trade stream had become metered so it delivered nothing. Live candles were also never re-fetched.
Expected behaviour	The candle updates within a second of a swap. The app now reads the token's pool reserves straight from Solana (push with a polling fallback), re-fetches closed candles every 30 seconds without resetting the chart, and the source line shows the pool type and the age of the last update.
Image proof	No capture yet

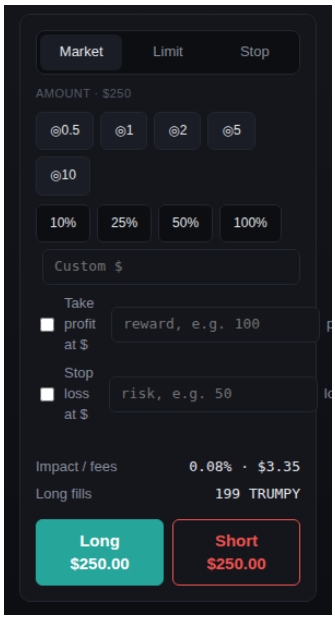
13. Phantom on the iPhone app went to Phantom and never came back	**Fixed, needs a live check**
Description	Connect opened the site inside Phantom's own browser, so the wallet was usable there but not inside the Expo app.
Expected behaviour	Connect opens Phantom with a deep link, the user approves, Phantom returns to the MemeScreen app and the wallet shows on the Portfolio page.
Image proof	No capture yet

14. Market summary arrow on the dashboard did nothing	**Fixed**
Description	The "Market summary" heading with the arrow was plain text. Clicking it did nothing.
Expected behaviour	It is a button. Clicking it opens the featured token in the Trade page.
Image proof	No capture yet

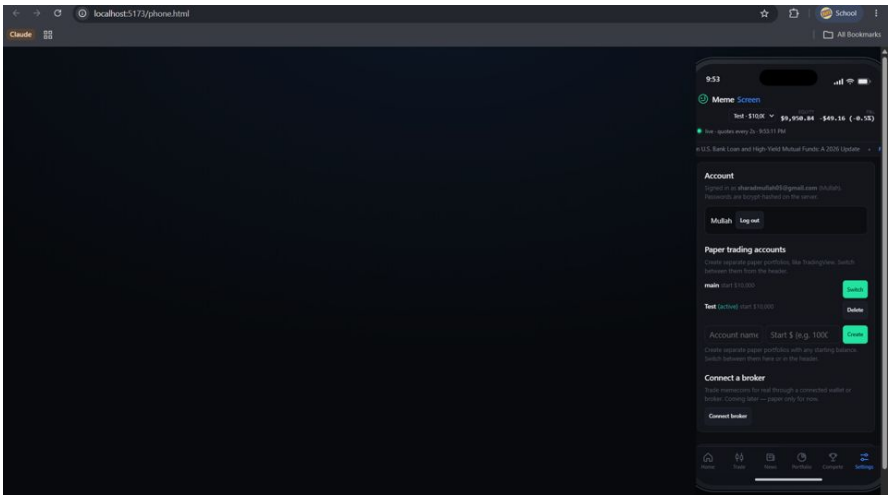
15. Joining a competition did not work	**Fixed**
Description	Competitions were saved on the device that created them. A code typed on another device or another login found nothing, so nobody could join.
Expected behaviour	Competitions live in Back4App. Create gives a code, anyone with the code joins, the competition has its own board of its members, and members can leave.
Image proof	

16. News links replaced the app	**Fixed**
Description	Tapping a headline inside the iPhone app loaded the news site in place of MemeScreen, with no way back. On the web the same click sometimes navigated the tab away.
Expected behaviour	Every outside link opens in a new tab on the web and in Safari on the phone. The iPhone shell refuses to load any page that is not MemeScreen.
Image proof	No capture yet

Fixed in earlier builds

17. Take profit and stop loss fields ran off the order panel	**Fixed**
Description	The two rows were wider than the 300 pixel panel. The labels wrapped and the words "profit" and "loss" were cut off at the edge.
Expected behaviour	Both rows fit inside the panel at every width, with the dollar field next to its checkbox.
Image proof	

18. Short positions got the wrong take profit and stop loss	**Fixed**
Description	Exit prices were always worked out as if the trade was a long. A short got a take profit above its entry and closed on the next price tick.
Expected behaviour	For a short, take profit sits below entry and stop loss above. Limit and stop orders measure from the trigger price.
Image proof	No capture yet

19. Phone simulator sat off-centre	**Fixed**
Description	On a wide window the phone rendered against the right edge and the controls were hidden.
Expected behaviour	The phone is pinned to the centre of the window at any size. Controls float at the side and can be hidden.
Image proof	

20. Chart and order ticket reset every 15 seconds	**Fixed**
Description	The Trade page rebuilt itself on a timer. Drawings were wiped and anything typed into the ticket was lost.
Expected behaviour	The chart, its drawings and the ticket contents survive every refresh.
Image proof	No capture yet

21. Take profit and stop loss lost on reload	**Fixed**
Description	Exit levels were not saved for signed-in users.
Expected behaviour	Positions come back after a reload with their exits.
Image proof	No capture yet

22. Market cap column showed the token price	**Fixed**
Description	The live update wrote the price into the market cap cell of the token list.
Expected behaviour	The column shows market cap and updates live.
Image proof	No capture yet

23. Blank error on wrong password or duplicate email	Fixed
Description	The message was cleared as soon as it was set.
Expected behaviour	"Wrong password", "No account with that email" or "That email is already registered" stays visible.
Image proof	No capture yet

18. Planned next

Item	What it means
Back4App permissions	Lock class-level permissions and turn off client class creation. Deploy the cloud function for the dashboard market tiles.
Hosting	Put the site on https so phones can install it and receive system notifications
Server push	Send alerts when the app is closed. The service worker already has the receiving side.
Store builds	Windows and macOS installers from Tauri. iOS and Android packages through a Capacitor or Tauri mobile wrapper.
Drawings on the account	Save drawings to the backend so they follow the login
RugCheck through the backend	Move the holder calls to a Back4App cloud function if the browser calls are blocked or rate-limited
Five-tab phone layout	Fold Portfolio, Compete and Settings under a More tab to match Apple guidance
Token sentiment panel	Bull or bear read on each token with analyst-style notes and links to the latest news
Indicator and strategy builder	Users write their own indicators and strategies, run them in an in-app terminal, and build strategy reports
AI strategy testing	Assisted back-testing of a user's strategy
Server-side alerts	A Back4App scheduled job that checks price rules and screens and calls pushToUser, so alerts arrive with the app closed
Live order history	Record live swaps in the Portfolio journal next to paper trades
Hardening before public launch	Email verification, server-side trade validation, security review

19. App flow

Step	Web	Phone
1	Landing page	Home screen, tap the MemeScreen icon
2	Log in, create account, or browse as guest	Sign-in screen: log in, create account, or guest
3	Dashboard	Home tab
4	Screener: pick a screen or write one, tap a token that matches	Screen tab: same, editor opens as a sheet
5	Trade: read the grade and flags, chart it	Trade, Chart and trade segment
6	Mark levels with the drawing tools	Same tools from the top toolbar
7	Set size, TP and SL, go long or short	Tap Trade, set size, TP and SL, go long or short
8	Watch the position in the bottom panel, close 25%, 50% or all	Positions segment, same buttons on each card
9	Alerts arrive as cards and system notifications	Alerts arrive as phone banners, in the app or on the home screen
10	Portfolio for the journal, Competitions for the board	Same, from the tab bar